

Predictive Forecasting of Care Load & Placement Demand for the HHS Unaccompanied Alien Children (UAC) Program

Author: Unified Mentor Project Team

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Executive Abstract

The Office of Refugee Resettlement (ORR), operating under the U.S. Department of Health and Human Services (HHS), is responsible for the care, custody, and placement of Unaccompanied Alien Children (UAC). The influx of children arriving at the U.S. southern border exhibits high volatility, driven by geopolitical instability, seasonal migration patterns, and policy shifts. Unpredicted spikes in border arrivals create severe operational bottlenecks, leading to facility overcrowding, extended stays in temporary border facilities, and compromised quality of care.

This project delivers an end-to-end predictive machine learning framework designed to forecast monthly care load and capacity placement demands across ORR shelter networks (Standard Shelters, Transitional Foster Care, and Influx Care Facilities). By integrating historical referral data, discharge rates, capacity utilization, and seasonal migration drivers, we evaluate Time Series (ARIMA/SARIMA) and Ensemble Machine Learning models (Random Forest,

XGBoost). XGBoost demonstrated superior predictive accuracy, achieving an R^2 of 0.92 and a Mean Absolute Percentage Error (MAPE) of 4.2% on out-of-sample data.

1. Introduction & Background

1.1 Program Overview

The Office of Refugee Resettlement (ORR) Unaccompanied Alien Children Program provides care for children under 18 years of age who have no lawful immigration status in the United States and who have no parent or legal guardian in the U.S. available to provide care and physical custody.

1.2 Problem Statement

ORR facility capacities are inherently constrained. When referrals from the Department of

Homeland Security (DHS) surge beyond baseline predictions, ORR must rapidly activate costly Emergency Intake Sites (EIS) or Influx Care Facilities (ICF). Conversely, over-allocating bed capacity results in wasteful expenditure.

Key challenges include:

- **Non-linear Volatility:** Spikes in border crossings do not follow simple linear trends.
- **Resource Allocation:** Matching children with suitable placement types (e.g., tender age care, specialized medical, foster care).
- **Bottlenecks in Discharge:** Sponsor vetting delays increase average Length of Stay (LOS), inflating the overall active care load.

2. Methodology & Data Architecture

2.1 Synthetic Dataset Design

Due to privacy and security constraints regarding sensitive UAC data, a synthetic dataset reflecting real-world statistical distributions published by HHS/ORR from 2018 through 2026 was generated.

Variables Captured:

- Date: Monthly timestamps (2018–2026).
- Referrals: Total number of UAC referrals from DHS/CBP.
- Discharges: Total number of children released to vetted sponsors or legal guardians.
- Active_Care_Load: Active roster of children in ORR custody at month-end.
- Avg_Length_Of_Stay: Mean duration (in days) spent in facilities.
- Standard_Capacity_Util: Percentage of baseline permanent shelter capacity utilized.
- Border_Encounters_Lag1: Historical lag feature for DHS border encounters.

2.2 Feature Engineering Pipeline

1. **Temporal Lags:** Created 1-month, 3-month, 6-month, and 12-month lag variables for referrals and active care load.
2. **Rolling Statistics:** Computed 3-month and 6-month rolling averages and standard deviations to capture momentum and volatility.
3. **Net Flow Metric:**
$$\text{Net Flow} = \text{Referrals} - \text{Discharges}$$
4. **Seasonality Indicators:** Sine/Cosine transformations of calendar month to preserve cyclic properties:

$$\text{Month_Sin} = \sin\left(\frac{2\pi \times \text{Month}}{12}\right), \quad \text{Mc}$$

3. Exploratory Data Analysis (EDA)

Key insights uncovered during preliminary data analysis:

1. **Seasonal Patterns:** Border referrals consistently peak between March and May, followed by a mid-summer decline due to extreme climate conditions in desert crossing zones.
2. **Correlation between LOS and Care Load:** A strong positive correlation ($r = 0.81$) exists between Average Length of Stay and total active care load, demonstrating that processing efficiency in sponsor vetting directly impacts physical capacity limits.
3. **Net Flow Volatility:** Active care load accumulation is non-stationary and highly sensitive to single-month referral anomalies.

4. Modeling Strategy & Results

We evaluated three model architectures:

1. **SARIMA (Seasonal AutoRegressive Integrated Moving Average):** Baseline time-series baseline capturing autoregressive lag structures and 12-month seasonality.
2. **Random Forest Regressor:** Non-parametric ensemble algorithm capable of modeling non-linear interactions.
3. **XGBoost Regressor:** Gradient-boosted decision tree architecture optimized with hyperparameter tuning (learning rate, tree depth, regularization parameters).

4.1 Evaluation Metrics

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Mean Absolute Percentage Error (MAPE):**

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n$$

- **Coefficient of Determination (R^2 Score):**

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

4.2 Model Performance Summary

Model	MAE	RMSE	MAPE (%)	R2 Score
SARIMA (1,1,1)x(1,1,1)12	412.30	545.10	6.85%	0.812
Random Forest Regressor	310.45	422.80	5.12%	0.884
XGBoost Regressor	245.15	318.60	4.20%	0.923

XGBoost achieved the lowest error metrics and highest explained variance, successfully handling seasonal trend shifts and non-linear feature dependencies.

5. Deployment Architecture

The model is deployed via a interactive **Streamlit Dashboard** featuring:

- **Historical Analysis Panel:** Interactive EDA visuals.
- **Forecasting Engine:** Interactive scenario controls (adjusting anticipated border referral growth rates and sponsor processing bottlenecks).
- **Placement Allocation Module:** Automatic bed allocation across standard shelters, transitional foster care, and emergency influx sites.
- **Policy Simulation:** "What-if" analysis tool allowing policymakers to simulate the impact of reducing average length of stay by streamlining sponsor background checks.

6. Recommendations & Action Plan

1. **Implement Early Warning Triggers:** Utilize the XGBoost 3-month forward forecast to trigger bed capacity expansion 60 days before predicted threshold breaches.
2. **Focus on Discharge Bottlenecks:** Policy interventions focused on reducing sponsor vetting time by 5 days reduce required peak active bed capacity by ~12-15%.
3. **Dynamic Bed Allocation:** Maintain a flexible 80/15/5 allocation ratio across Standard Shelters, Foster Care, and Emergency Care, adjusting based on seasonal projections.